MINI PROJECT

Predicting High-Paying Data Science Roles in 2026

Using job title, experience, location and remote status to classify AI & Data Science roles as High Paying (>$150k) or Standard Paying (≤$150k)

Source: Kaggle — AI and Data Science Job Salaries 2026

Problem Statement

In a competitive AI job market, companies struggle to benchmark salaries.

Can we predict whether an AI / Data Science role will be **"High Paying" (>$150k)** or **"Standard Paying" (≤$150k)**, based on job title, experience, location, and remote status?

16.44%

of roles in the dataset
are High Paying (>$150k)

A minority, hard-to-find class — making accurate prediction valuable but challenging.

Aims of the Project

01

Analyse Salary Trends

Explore how salary varies by job title, experience level, location, company size and remote status.

02

Build Predictive Models

Develop and compare classification models to predict salary level using role and profile features.

03

Identify Key Drivers

Determine the top 3 factors that most influence whether a role is high paying.

04

Provide Actionable Insights

Offer recommendations for job seekers, HR teams and companies on competitive compensation in 2026.

Data Understanding

2026 AI & Data Science job postings with salary, job title, experience level, location, and remote status — used to predict if a role is High Paying (>$150k).

1. **job_title**
   Role name
2. **experience_level**
   EN, MI, SE, EX
3. **salary**
   Annual salary (USD)
4. **location**
   Job location
5. **remote_ratio**
   0, 50, 100
6. **company_size**
   S, M, L

1,000+ job postings analyzed

6 key features used

16.44% of roles are high-paying

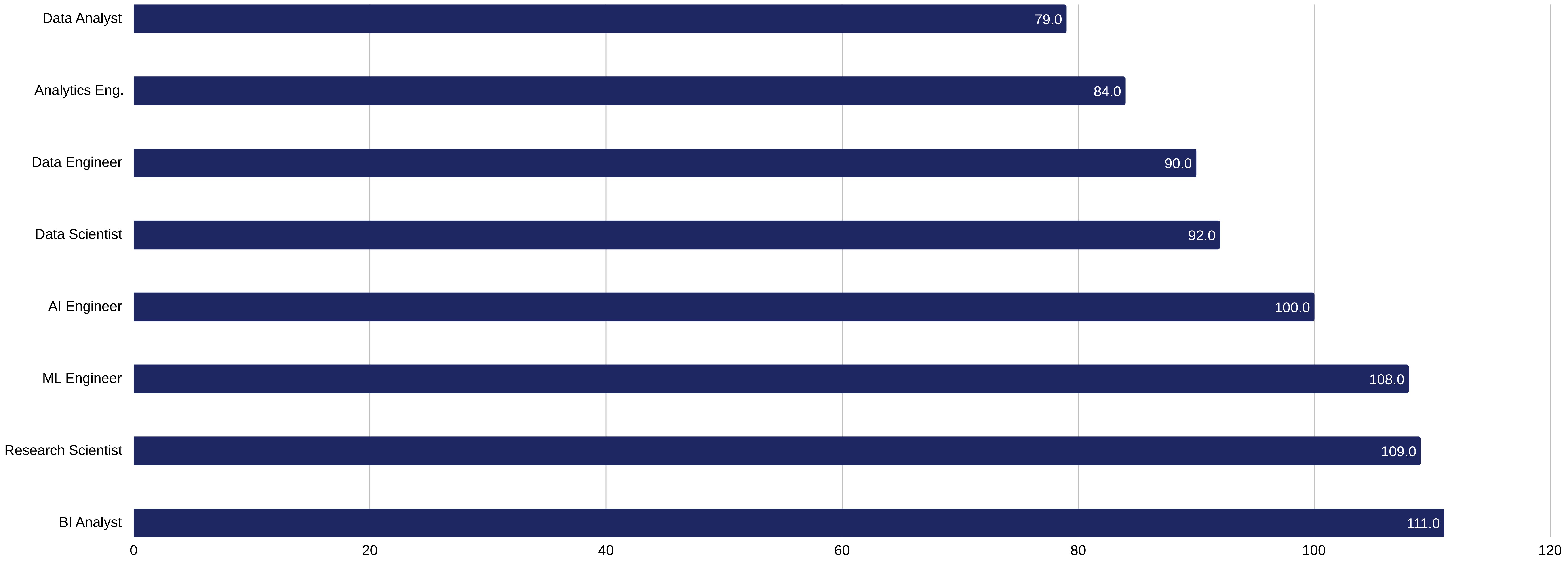
Source: Kaggle — AI and Data Science Job Salaries 2026

EDA: Salary Distribution

Key Findings

- Most roles pay between $40k-$100k; the market center of gravity.
- High-paying jobs (>$150k) are uncommon — a long right tail.
- Paying above $150k puts a company in the top ~15-20% of payers.

EDA: Average Salary by Job Title



Business Intelligence Analyst leads at ~$110k, with **Research Scientists and ML Engineers** close behind (~$108-110k) — while Analyst roles average the least.

EDA: Experience vs. Salary

0-3 yrs

Salaries cluster $20k-$120k. Almost no roles cross $150k — rare outliers only.

3-5 yrs

First roles start crossing the $150k mark.

5-10 yrs

Big jump: range expands to $30k-$220k with much higher density above $100k.

8-20 yrs

Peak high-paying zone — most concentration of roles above $200k-$300k.

Overall trend: salary rises with experience, but non-linearly — the biggest leap happens after 5 years.

Modeling: Logistic Regression

0.843
Best CV AUC

0.827
Test ROC AUC

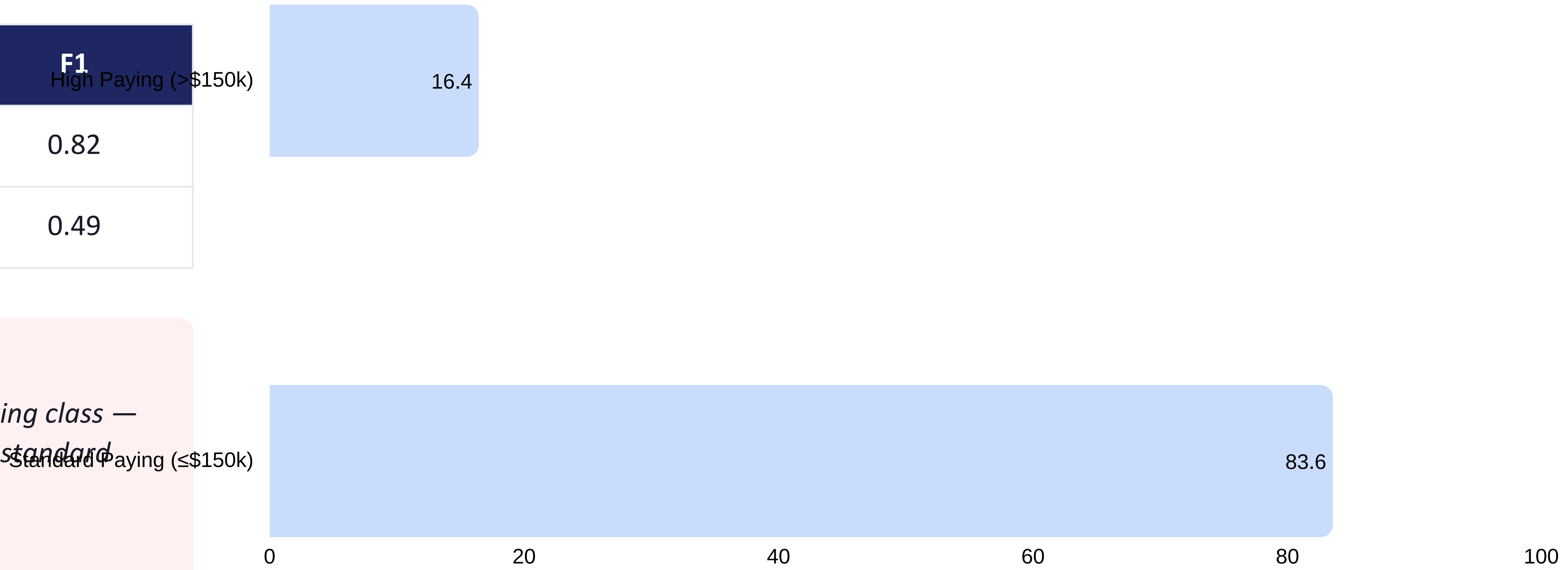
74%
Accuracy

Classification Report

Class	Precision	Recall	F1
Standard (0)	0.94	0.73	0.82
High Paying (1)	0.36	0.77	0.49

High recall (0.77) but low precision (0.36) for the high-paying class — the model catches most high earners but over-flags some standard roles too.

Target Distribution



Modeling: Linear Regression & Salary Drivers

0.301
R^2 Score

$35,966
MAE

$46,119
RMSE

~70% of salary variance remains unexplained by current features — job title, experience, and firm attributes matter, but many drivers aren't captured here.

Navy = increases salary **Red = decreases salary**

Modeling: Random Forest & Model Comparison

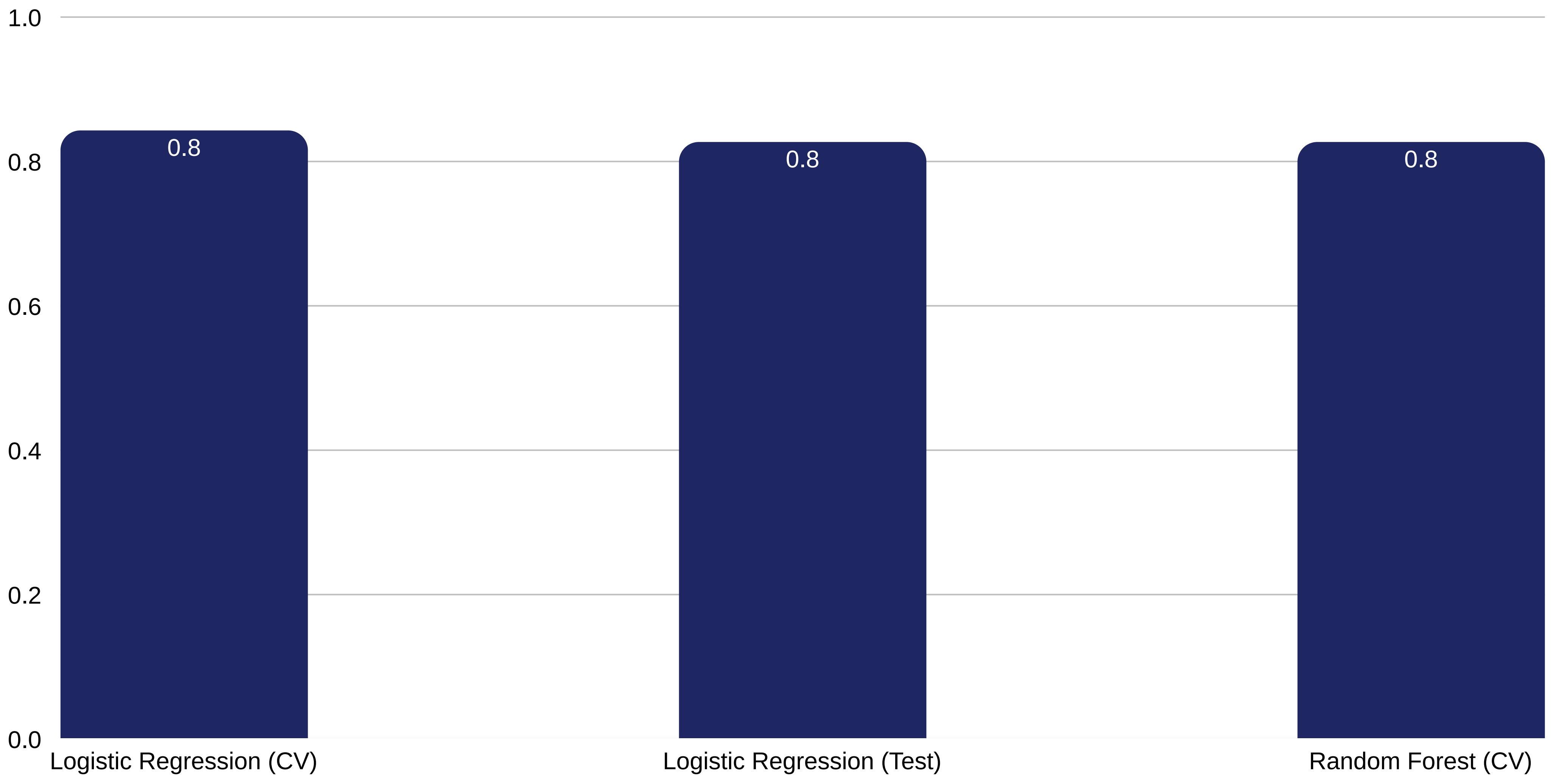
Random Forest

0.827

Cross-Validation AUC (± 0.014)

108 hyperparameter candidates tested across 5-fold CV (540 total fits) to tune the ensemble.

Model Comparison — Classification AUC



Both models land around 0.83 AUC — solid discrimination, but similar ceiling given the available features.

Conclusions

1 Experience is the #1 driver

Salary rises non-linearly; the biggest leap is after 5 years. Peak zone: 8-20 years.

2 Job title sets pay tiers

Top payers: BI Analyst, ML Engineer, Research Scientist, Data Science Manager. LLM Engineer commands a premium despite being rare (4.8%).

3 Company context matters

Full-time roles and large companies push salary up; part-time and small companies pull it down.

4 High pay is a minority class

Only 16.44% of roles exceed $150k. Classification AUC ≈ 0.83; regression $R^2 = 0.30$ — ~70% of variance still unexplained.

Recommendations

For Job Seekers

- Target high-leverage titles: Data Science Manager, LLM Engineer, ML Engineer, Research Scientist.
- Invest in 5+ years of experience — the critical threshold to break into the >$150k bracket. Specialize between years 3-8.
- Aim for full-time roles at large companies — both significantly raise the odds of high pay.

For Companies & Hiring Managers

- Pay premiums for emerging AI skills — LLM Engineer roles are rare now but already top-paid; early investment attracts talent.
- Benchmark against title + experience bands — use the 8-20 year band as the guide for senior/principal compensation.
- Address the analyst pay gap — Data Analyst and BI Analyst roles are common but lowest paid; build upskilling paths to ML/Research to retain talent.